7 **(a)** State *Newton's second law of motion*.

.....

.....

..... [2]

(b) A rollerblader approaches and glides up a smooth ramp of length 2.0 m along AB, inclined at an angle of 30° to the horizontal. The side view of this ramp is shown in Figure 7.1.

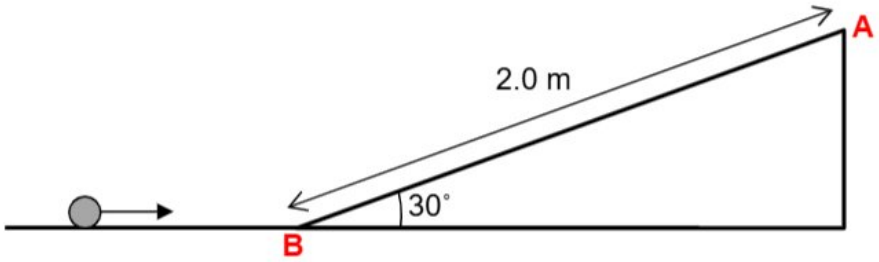


Figure 7.1

Figure 7.2 shows the ramp when viewed **perpendicular** to the ramp's inclined surface. The rollerblader glides up the ramp with initial speed u, at an angle of $\theta = 60°$ to the base of the ramp as shown in Figure 7.2. He stops exerting energy the moment he starts moving up the ramp.

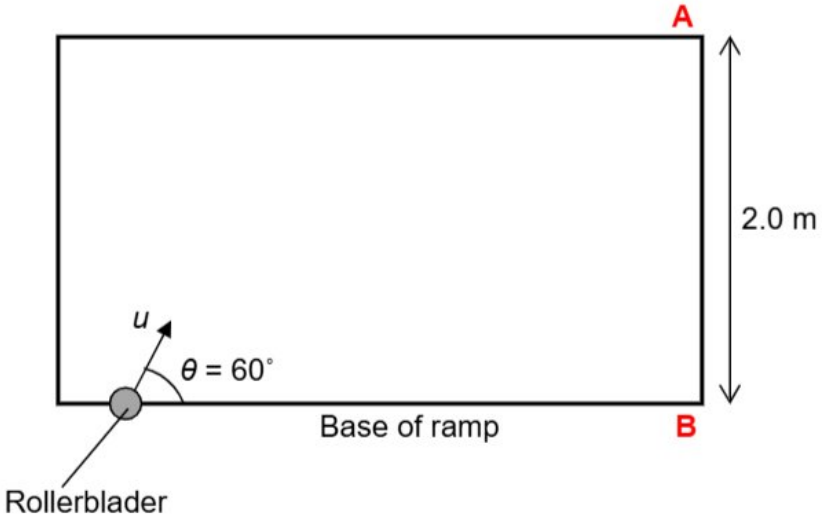


Figure 7.2

(i) Determine the magnitude of the acceleration of the rollerblader while moving up the ramp.

acceleration = ……..…….m s^{-2} [2]

(ii) On Figure 7.2, draw an arrow to show the direction of the acceleration of the rollerblader. [1]

(iii) Explain why the rollerblader does not move up the ramp along a straight path but along a parabolic path.

……

……

……

……

……

…… [3]

(iv) By using the principle of conservation of energy, determine the minimum speed *u* for the rollerblader to just make it to the top of the ramp.

minimum speed = ……..…….m s^{-1} [3]

(v) Using the speed in **(b)(iv)**, determine the time taken for the rollerblader to return to the base of the ramp.

time taken = ...s [2]

(c) The ramp is extended to allow the rollerblader to move through a partial vertical circular motion with radius 3.0 m. Figure 7.3 shows the side view of the entire ramp. The rollerblader is upside-down at the top of the ramp.

The rollerblader now approaches the base of the ramp straight on perpendicularly.

You may regard the rollerblader as a point mass throughout this motion.

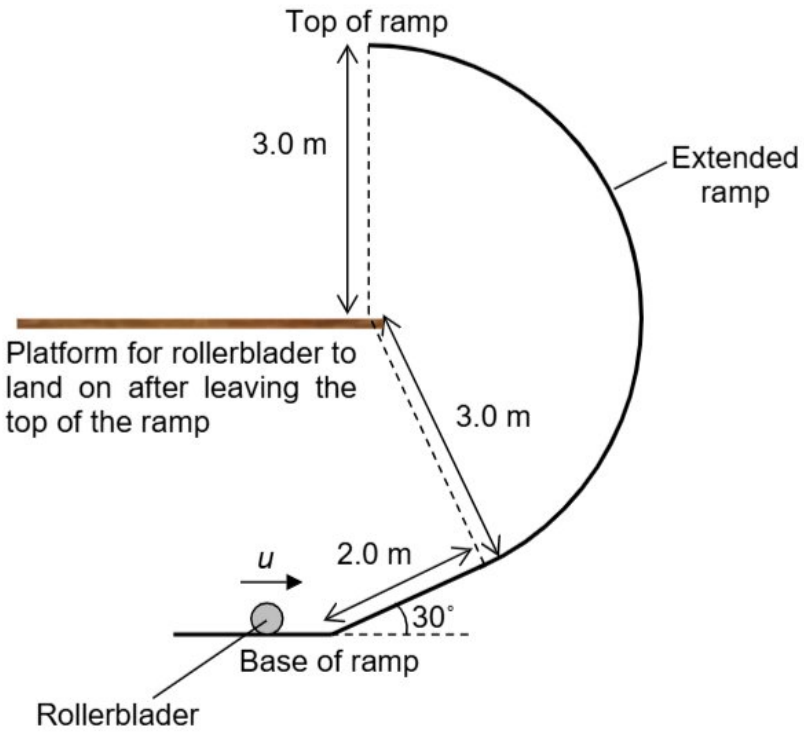


Figure 7.3 (not to scale)

(i) On Figure 7.4 below, sketch a graph to show how the magnitude of the centripetal acceleration a_c of the rollerblader varies with his linear speed v.



Figure 7.4 [1]

(ii) Determine the minimum speed u required for the rollerblader to reach the top of the circular ramp shown in Figure 7.3.

minimum speed = ...m s^{-1} [4]

(iii) On Figure 7.5 below, sketch a graph to show how the kinetic energy of the rollerblader E_K varies with his vertical displacement above the base of the ramp s_y when he moves up the ramp with this minimum speed.

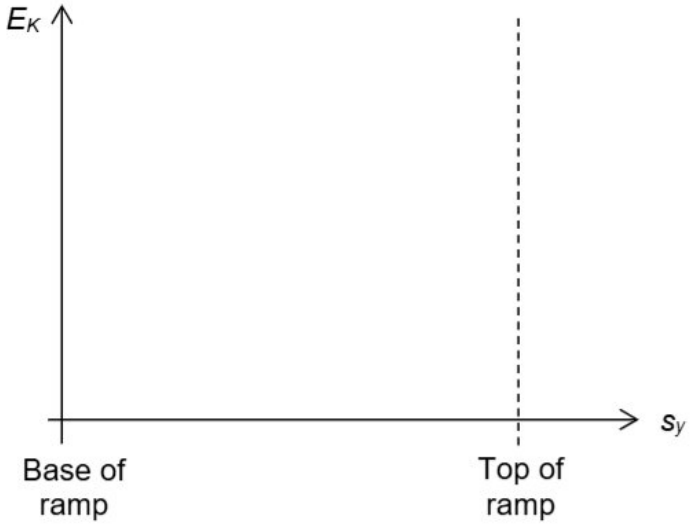


Figure 7.5 [2]

[Total: 20]